

CT-268 PROFESSIONAL ETHICS

LECTURE 6

ETHICAL PROBLEM-SOLVING TECHNIQUES: LINE DRAWING AND FLOWCHARTING FOR CONFLICT PROBLEMS

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CONFLICT PROBLEMS

- Conflict problems are a clash between competing **obligations, interests, or rules.**
- Conflict problems and ethical dilemmas are often used interchangeably in casual conversation, but they have distinct structural definitions in engineering and computing ethics frameworks.
 - Remember that ethical dilemma is a choice between two or more equally valid moral principles where choosing one inevitably violates the other.
- Conflict problems are highly resolvable using systematic frameworks, priority rankings, or compromise; whereas, ethical dilemmas often leave a "moral residue" (no matter what you choose, a deep moral principle is broken.)



Three Classic Types of Conflict Problems

- **Obligation vs. Obligation (The Multi-Duty Clash):**
 - This happens when two core clauses of a professional code of ethics (like the ACM or IEEE codes) slam into each other.
- **Professional Duty vs. Personal Loyalty/Interests:**
 - This occurs when a professional's objective, neutral judgment is compromised by personal relationships, financial gain, or career advancement.
- **Corporate Loyalty vs. Whistleblowing:**
 - This is the most common conflict problem taught in professional ethics classes. It pits organizational hierarchy against public transparency.
- In this lecture we shall discuss two methods to resolve conflict problems:
 - Line-Drawing
 - Flowcharting



LINE-DRAWING

- Where to set the boundary in ethical problems.
- Although definitions of concepts are open-ended, this does not mean that every application of a concept is problematic.
- In fact, it is usually quite easy to find clear-cut, unproblematic instances.
- We can refer to these as paradigm cases.
 - Positive Paradigm and Negative Paradigm.
- Steps:
 - Identify features of the paradigm case.
 - List the features and its paradigm instances.
 - Perform the Line-Drawing Test of Concepts.
 - Perform the Line-Drawing for Paradigm and Features.



Consider the Scenario

Suppose Amna signs an agreement with Company A (with no time limit) that obligates her not to reveal its trade secrets. Amna later moves to Company B, where she finds a use for some ideas that she conceived while at Company A. She never developed the ideas into an industrial process at Company A, and Company B is not in competition with Company A, but she still wonders whether using those ideas at Company B is a violation of the agreement she had with Company A. She has an uneasy feeling that she is in a gray area and wonders where to draw the line between the legitimate and illegitimate use of knowledge. How should she proceed?



Let's Explore

- Because she feels she is in a gray area, it may be useful for her to compare her circumstance with a negative and positive paradigm in regard to taking one's ideas to a new place of employment.
- In determining what these paradigms might be, she should try to construct a list of key features that can be placed on a spectrum ranging from negative to positive.
- For example, violating a trade secret policy would be a negative feature, counting strongly against the appropriateness of taking her ideas to Company B.
- Acquiring permission from Company A would be a positive feature, counting strongly in favor of the appropriateness of taking her ideas to Company B.



Positive and Negative Features

- A case dominated by negative features would be a negative paradigm, a clear instance of wrongdoing.
- A case dominated by positive features would be a positive paradigm, a clear instance of acceptable behavior.
- Amna's situation is the test case.
 - Once Amna identifies the key features of her negative and positive paradigms, she can begin comparing the features of her situation with those of the paradigms.
 - For example, a negative feature of her situation is that she signed a trade secret agreement that may include her ideas and apparently she has not sought permission from Company A to use her ideas at Company B.
 - A positive feature is that Company A and Company B are not competitors.



Feature Analysis

- As Amna engages in this comparative analysis, she may find that she has not thought thoroughly enough about certain features.
- For example, she may not have thought much about the extent to which others at Company A might also have helped develop her ideas.
- Or, although she developed her ideas on her own time, she might realize that Company A's lab and equipment played a crucial role in their development.
- Or, although Company A and B were not competitors when Amna worked at A, they might become competitors in the area in which she developed her ideas, especially if those ideas were jointly developed with others at Company A.



Further Investigation

- At this point, although Amna may feel she has a clearer understanding of her situation, she may still be unsure what to conclude.
- Some features of her case lean in the direction of features of the negative paradigm, whereas others lean in the direction of the positive paradigm.
- Furthermore, in this particular case some of the negative and positive features may be more important than others and should be more heavily weighted.
- Therefore, Amna still has to assess the importance of the various negative and positive features she is considering.
- She may think of other possible scenarios that fall somewhere between the negative and positive paradigms, and she can compare the features of her case with those of the intermediate cases



Features of Paradigms

Negative Paradigm (Clearly Wrong)	Positive Paradigm (Clearly Acceptable)
Signed agreement	Permission granted
A and B competitors	A and B not competitors
Ideas jointly developed	Amna's ideas only
All ideas developed on job	All ideas developed off job
Heavy use of A's lab/equipment	A's lab/equipment not used



Paradigm Features and Test Cases

Negative Paradigm (Clearly Wrong)	Test Case	Positive Paradigm (Clearly Acceptable)
Signed agreement	--X-----	Permission granted
A and B competitors	-----X--	A and B not competitors
Ideas jointly developed	-----X----	Amna's ideas only
All ideas developed on job	-----X---	All ideas developed off job
Heavy use of A's lab/equipment	-----X-----	A's lab/equipment not used



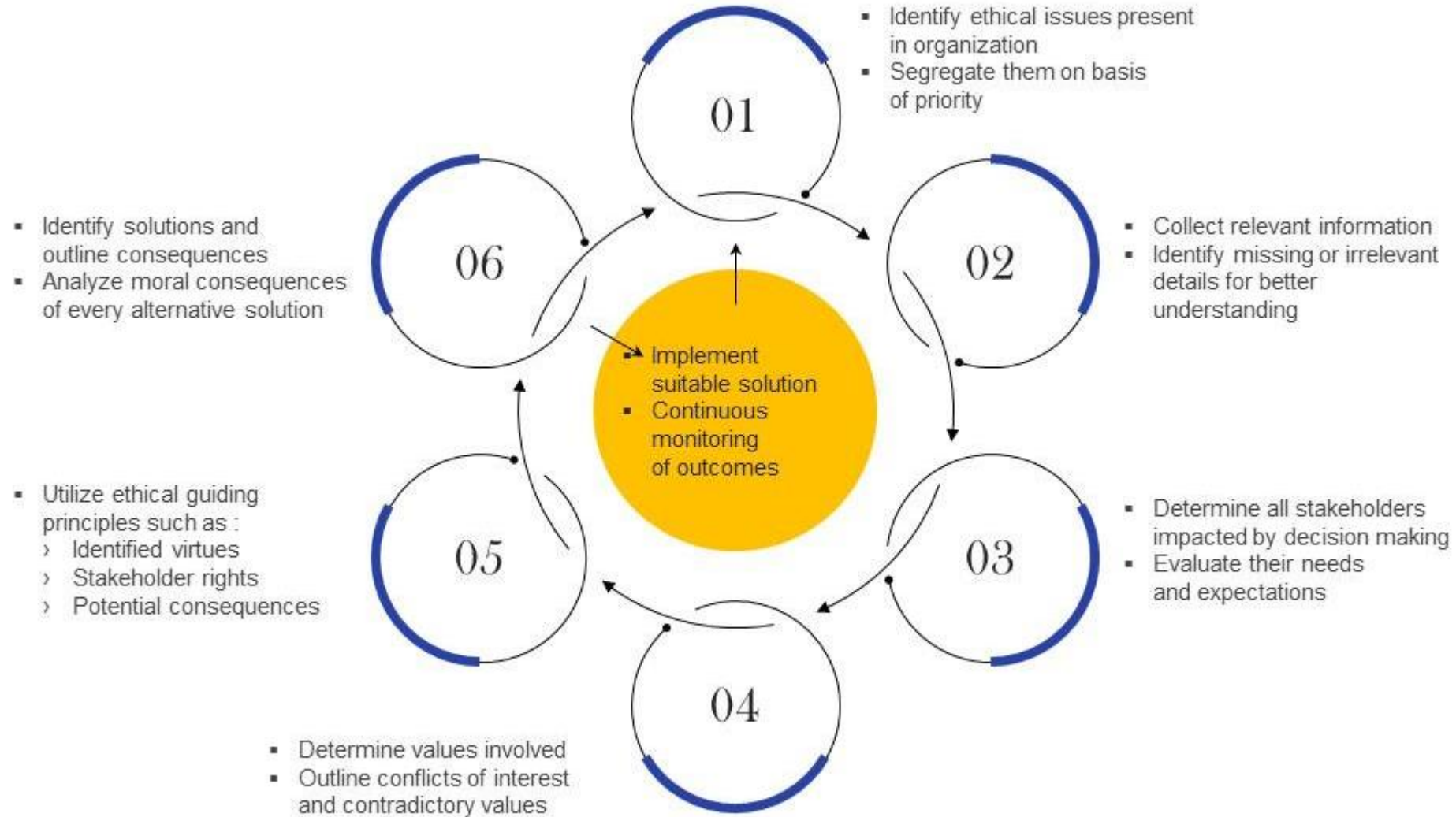
Test Case

- Those two or three features that are judged most important in a particular case can be identified by drawing a circle around the appropriate X's.
- Cases that are certainly wrong we shall call negative paradigm cases, and cases that are certainly acceptable are positive paradigm cases.
- We shall call related, but controversial, cases that are in dispute (and that are clustered near the middle of the spectrum) problematic cases.
- We shall designate as the **test case**, the one on which the analysis is to focus.

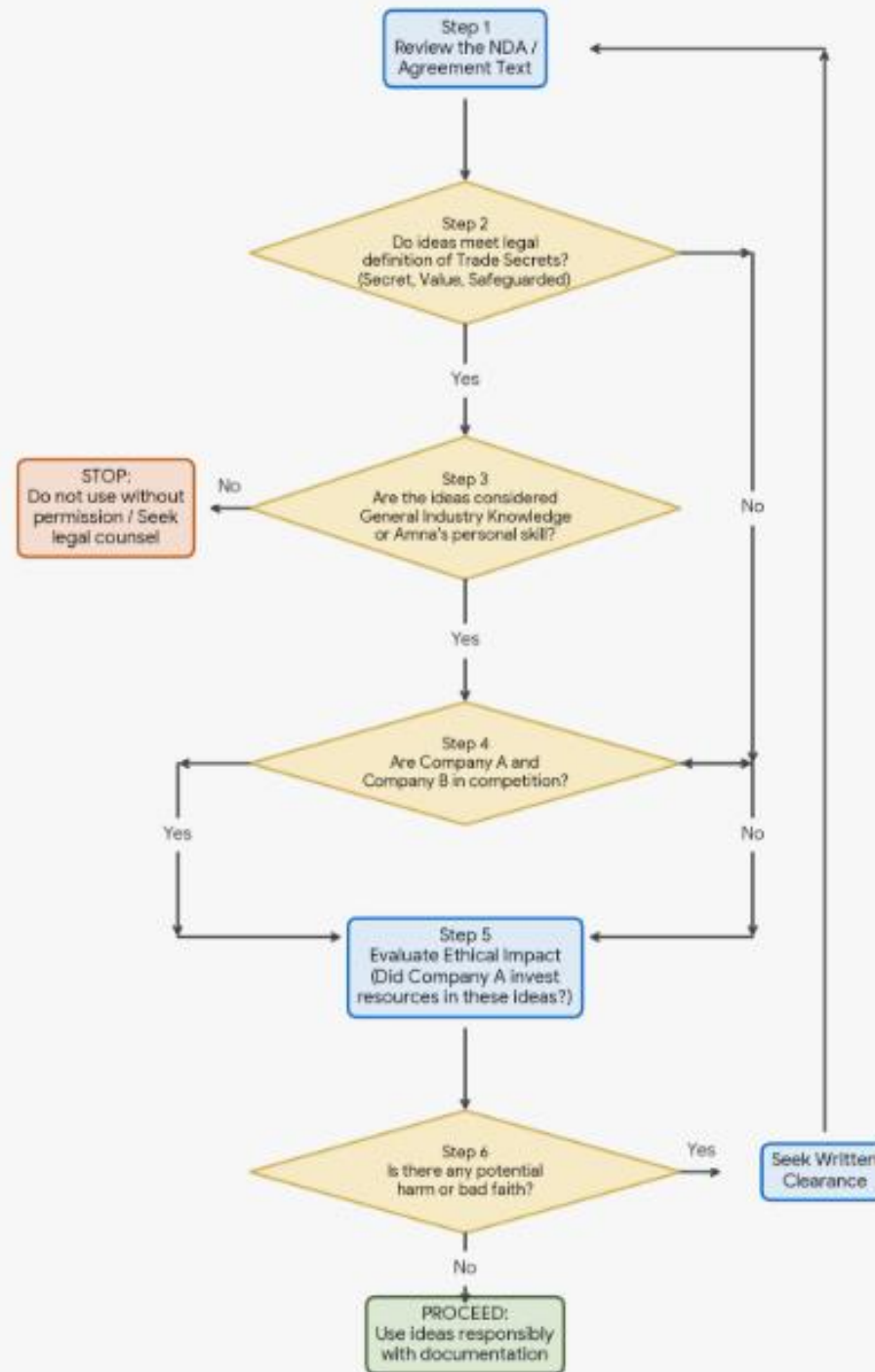


FLOWCHARTING

- A flowcharting method for ethical decision-making provides a visual, step-by-step path to untangle complex dilemmas, weigh competing values, and justify a course of action.
- Steps in Ethical Flowchart:
 - Spot the trigger event and confirm whether an actual ethical or legal problem exists.
 - Collect first-hand data, identify stakeholders, and map out who, what, when, where, and why without rushing.
 - Check relevant laws, organizational guidelines, professional codes of ethics, or conflicts of interest.
 - List multiple potential courses of action; if you only see two polarized choices, look for a blended compromise.
 - Test options using major ethical/moral frameworks/theories.
 - Select the best option while communicating with transparency, then document the rationale and execute the decision.
 - Assess the outcomes after implementation and note what can be learned or handled differently next time.



For the Previous Example



Exercise 1



Qasim is a machine learning engineer working for a healthcare app that assists triaging emergency room patients. When the AI model was trained two years ago, it passed all rigorous safety benchmarks for identifying high-risk cardiovascular symptoms. However, during a routine monitoring check, Qasim notices model drift—the algorithm's accuracy for a specific minority demographic has silently dropped by 7% over the last six months due to changing patient demographics in the city. Qasim alerts his immediate product manager. The manager tells him that taking the model offline to retrain it will cause a massive operational bottleneck for hospitals, damage the company's reputation, and trigger an expensive regulatory review. The manager instructs him to quietly work on a patch for the next major software release, which is six months away. The app is technically still highly effective for the vast majority of users, and pulling it could cause immediate chaos in ER waiting rooms. However, leaving it active means a small percentage of patients are currently facing a higher risk of misdiagnosis. Is it ethical for Qasim to stay silent while working on the slow patch, or should he escalate the issue externally?

Exercise 2



Azam is a DevOps engineer at a cloud-based project management SaaS provider. The company is actively in talks to be acquired by a massive tech conglomerate, a move that would net all early employees life-changing payouts. During a security audit of the infrastructure, Azam discovers that a customer telemetry tool (intended strictly to track system crashes) is accidentally capturing and storing plaintext corporate passwords and sensitive user notes from several high-profile clients in an unencrypted log file. Azam brings this to the Chief Technology Officer (CTO). The CTO realizes that reporting this data breach will legally require notifying the affected clients, triggering public disclosure laws. This disclosure would almost certainly kill the acquisition deal and ruin the company's valuation. The CTO orders Azam to silently patch the logging tool immediately and securely delete the historical unencrypted log files, arguing that because no external hackers have accessed the file, "no actual harm was done." Deleting the logs removes the immediate security vulnerability, but it completely covers up the fact that client data was exposed in plaintext for months. By obeying, Azam protects his coworkers' livelihoods and his own financial windfall. By refusing or whistleblowing, he protects consumer privacy rights but destroys his own career prospects at the firm. How should he proceed?

